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Director of Product Management
Cookbooks & Developer Experience Team
ArangoDB Inc.
San Francisco, CA, USA

Dear Director of Product Management and the Arango Engineering Team,

ArangoDB's mission to eliminate fragmented AI stacks by unifying graph, vector, document, and search capabilities on a single contextual data layer transforms how enterprise AI agents access trusted business context. Operationalizing this multi-model architecture requires practical, turn-key recipes in Python and Jupyter Notebooks showcasing clear business value and reproducible engineering patterns. Having built and benchmarked open-source retrieval pipelines in **Python**, **Jupyter Notebooks**, and **ChromaDB**, I have tackled the exact tradeoffs between dense vector search, hybrid ranking, and contextual document ingestion that developers encounter daily.

In my open-source retrieval benchmark **ncert-rag-evals**, I engineered an empirical evaluation suite across 17 textbooks in **Python** and **Jupyter Notebooks**, implementing custom hybrid search from scratch in **SQLite FTS5** and **ChromaDB** to compare retrieval strategies against raw CPU latency. By combining BM25 lexical signals with dense vector embeddings via Reciprocal Rank Fusion (RRF), I proved an empirical **86.9% Recall@5 in 8.9ms on CPU**, cutting latency by **99.8%** versus LLM query expansion. Furthermore, in **vectorless-rag**, I conducted interactive retrieval analysis in **Jupyter Notebooks** using **Matplotlib** to benchmark generative page indexing against dense vector RAG across 18 queries, plotting cost-latency trade-offs while streaming document windows via **PyMuPDF** under a strict **under-200MB memory ceiling**.

Production recipes must adhere to rigorous software engineering and evaluation discipline rather than functioning as fragile script prototypes. During my engineering internship at **LexoLegal**, I architected asynchronous REST APIs in **FastAPI** and **PostgreSQL** behind an **Nginx** proxy, integrating **Model Context Protocol (MCP)** clients and **Celery** workers with atomic **Redis** distributed locks to handle burst traffic with **sub-2s latency**. Across all my open-source repositories, every workflow is validated with extensive test suites (including 92 **Pytest** suites in **ncert-rag-evals** and 151 tests in **doubtless**), ensuring that recipes published to the Arango Cookbooks repository will run deterministically for every developer.

I am eager to contribute 4 high-impact, reproducible recipes to the Arango Cookbooks repository as a Software Engineer Intern. As a Computer Science undergraduate based in India, I am available for immediate engagement within the India cohort, fully prepared for remote collaboration across distributed timezones, and welcome the opportunity to discuss recipe concepts with your team.

Sincerely,

Aman Verma